

What the AI Actually Costs to Run

— every agent, every cycle, at group scale

A transparent, bottom-up estimate of the AI (token) cost to run Aerchain's autonomous procurement spine — from intake capture all the way to invoice match — across ~300,000 procurement cycles a year. Costed agent-by-agent, by deal complexity. A directional planning model, not a contractual quote.

ANNUAL VOLUME MODELLED	DISTINCT AI AGENTS	ESTIMATED AI COST	BLENDED PER CYCLE
~300,000 cycles / year	36 across intake + pay	~\$804K / year : ~\$67K / month	~\$2.68 / procurement cycle

Prepared for **Adani Group** — IT & Procurement — companion to the Aerchain + Adani Commercial Proposal

Directional model at current AI pricing

1 How We Estimate This — and What's in Scope

A fair question deserves a clear answer: "how do I understand the costing?" This is the honest, bottom-up answer — not a flat per-seat number, but the actual cost of the AI work each procurement cycle triggers. This is the **internal fixed cost** that sits underneath the Aerchain platform subscription.

THE MODEL IN ONE LINE

Cost = the **AI work** each agent performs, summed across every agent in a cycle, multiplied by **how complex** the cycle is, multiplied by **how many** cycles you run a year. Nothing else is in this number.

WHAT'S COUNTED

AI processing only

The direct cost of the AI engines (tokens) running every agent across the cycle, at current rates.

WHAT'S NOT COUNTED

Platform & infra are separate

Hosting, database, search, document storage, and the Aerchain platform subscription are not in this figure. This isolates the AI line so it can be reasoned about on its own.

SCOPE OF THE CYCLE

Intake → award → pay

The full autonomous spine: intake capture, RFx drafting, offer extraction, negotiation, evaluation, award — and the downstream invoice extraction & match, plus the always-on assistant agents.

⚠ **A transparency note that holds for this whole document**

These figures are based on **current AI pricing**. AI processing costs have trended down meaningfully year over year, but they can change in either direction, and the agent mix evolves as the product matures. Treat this as a directional planning model that we will re-baseline against live usage — not a fixed price.

2 Every AI Agent in the Loop

A comprehensive map of the agents that run across a procurement cycle — 36 named agents, grouped by stage. Aerchain runs a **combination of AI engines**, routed per task across three tiers, so each job uses exactly the capability it needs and no more.

DEEP

Frontier reasoning

Reserved for the highest-stakes work — per-vendor technical & commercial scoring, award scenarios, negotiation strategy. Used only where it changes the outcome.

BALANCED

The everyday workhorse

Drafting, extraction and evaluation across most of the cycle — the best blend of quality and efficiency.

FAST

High-throughput

Lightweight engines for classification, ranking, routing and high-volume extraction — instant and economical.

Right-sizing the engine to the task — **deep only where it matters** — is a core reason the cost-per-cycle stays low.

AGENT	WHAT IT DOES	ENGINE
Intake Extraction	Pulls structured lines out of quotes, requirement docs, emails and uploads	Balanced
Vendor Resolver	Identifies the issuing vendor from a document	Fast
Enrichment	Normalises extracted lines against item / UOM / currency masters	Fast
Matching	Finds catalog & vendor candidates for each line	Fast
Reranker	Re-ranks candidates by semantic similarity	Fast
Routing	Picks the channel: PO - catalog - source - negotiate - buyer	Fast
Commercial-Rules Judge	Applies the buyer's own rules to fine-tune routing	Fast
Consolidation	Groups lines (vendor / category / location) for bundling	Fast

AGENT	WHAT IT DOES	ENGINE
RFx Generator	Drafts the full RFx — SOW, T&C, questionnaire, documents, vendor list	Balanced
RFx Section Generators	Generate individual sections (terms, item table, criteria)	Balanced
Attachment Extractor	Reads text & visuals from PDFs, images and documents	Balanced
Email Offer Extractor	Parses vendor offers from email / Excel / PDF into structured bids	Balanced
Vendor Follow-up	Chases vendors on a cadence until they respond	Fast

AGENT	WHAT IT DOES	ENGINE
Document / attachment analysis	Scores vendor documents for compliance & findings	Deep
Company-profile analysis	Vendor strengths, weaknesses, risk read	Balanced
Questionnaire scoring	Scores each vendor's questionnaire responses	Balanced
Contract & commercial-terms extraction	Pulls and normalises submitted terms per vendor	Balanced
Commercial normalization & summary	Normalises prices & terms onto a like-for-like grid	Balanced
Technical supplier recommendation	Synthesises a technical winner with rationale	Deep
Commercial recommendation	Synthesises a commercial winner with rationale	Deep

This stage alone runs ~15 AI passes per complex deal — most of them per-vendor — which is why complex cycles cost what they do.

AGENT	WHAT IT DOES	ENGINE
Target / Counter Generator	Extracts targets and counter-offers, sets floor / ceiling	Balanced
Playbook Generator	Builds the strategy: discovery → benchmark → pressure → BAFO	Deep
Vendor-message Draft	Drafts each buyer-side negotiation message	Balanced
Chat Runner	Runs multi-turn vendor conversations through the rounds	Balanced
Market-Benchmark Agent	Searches the market for defensible benchmarks (round 2+)	Balanced
Log Summary	Summarises outcomes & strategy effectiveness	Fast

A deliberate cost asymmetry — autonomy is highest where deals are simplest

Tail and standard deals run negotiation **autonomously** (the agent drafts and sends every round). Tactical and complex deals are **buyer-led** — the AI assists, a human drives. So negotiation cost actually *peaks at standard* and tapers on complex, where the spend shifts into deep evaluation instead. This is built into the per-cycle numbers.

AGENT	WHAT IT DOES	ENGINE
Scenario Constraint-Check	Feasibility pre-check before scenarios are built	Fast
Scenario Builder	Generates optimised award scenarios to compare	Deep
Scenario Verification	Validates generated scenarios against constraints	Balanced
Award Recommendation	Recommends the winner with rationale & guardrail check	Deep
Variance / Decision Gate	Auto-award vs. send-to-buyer against thresholds	Fast

AGENT	WHAT IT DOES	ENGINE
Invoice Extraction & Matching	Reads the invoice, 2-way / 3-way matches it against PO & goods-receipt, flags exceptions	Balanced

Because invoices outnumber purchase orders, this agent runs on a slightly larger base (~1.2x per cycle), already reflected below.

AGENT	WHAT IT DOES	ENGINE
Workbench / Help Agent	Answers any question, anywhere — queue triage, "what's stuck", how-to	Balanced
Analyst Agent (Aerchain Insights)	Read-only copilot on the RFx / analysis pages — compares bids, explains scores	Balanced
Draft-instructions Compiler	Turns rough buyer notes into clean agent instructions	Fast
Search Indexer	Indexes intake lines for semantic Q&A	Fast

The assistant agents are **usage-driven**, not per-transaction. We amortise them at an assumed ~2 help questions + ~1.5 analyst questions per cycle. Heavier adoption raises this line; lighter adoption lowers it.

3 The Four Complexity Tiers

Every cycle runs essentially the same agents — what changes is *how much* each one has to do: line items, vendors, questionnaires, documents, negotiation rounds. We band cycles into four tiers.

TAIL

Simple, high-volume

1-3 lines, 2-3 vendors, no real negotiation, lowest-price award. Mostly MRO / operational / repeat buys. Runs fully autonomously on the fast tier.

STANDARD

Routine sourced

5-15 lines, 3-4 vendors, 1 negotiation round, light commercial evaluation. The autonomous-negotiation sweet spot.

TACTICAL

More moving parts

15-40 lines, 5-6 vendors, 2 rounds, commercial + light technical evaluation. AI assists; a buyer drives the negotiation.

COMPLEX

Full techno-commercial

40-100+ lines, 10-12 vendors, full technical + commercial scoring, multiple award scenarios, Human-led negotiation; AI is the analysis engine.

4 Cost Per Cycle — by Stage and Tier

The AI cost of one complete intake-to-pay cycle, broken down by stage cluster. Figures are at current pricing, rounded.

STAGE CLUSTER	TAIL	STANDARD	TACTICAL	COMPLEX
Intake — capture, route & consolidate	\$0.04	\$0.08	\$0.12	\$0.20
RFx generation	\$0.10	\$0.22	\$0.58	\$1.06
Offer extraction (email / PDF / Excel)	\$0.04	\$0.12	\$0.24	\$0.60
Sourcing & vendor follow-up	\$0.02	\$0.04	\$0.06	\$0.12
Negotiation (playbook + target + rounds)	\$0.20	\$1.22	\$1.26	\$2.04
Evaluation & analysis (technical + commercial)	\$0.12	\$0.36	\$0.60	\$5.20
Award (scenarios + recommendation)	\$0.14	\$0.48	\$0.99	\$1.70
Invoice extraction & matching	\$0.08	\$0.16	\$0.24	\$0.50
Always-on assistants (help + analyst)	\$0.20	\$0.36	\$0.56	\$0.84
Total per cycle	\$0.94	\$3.04	\$4.56	\$12.26

Read the complex column

A complex cycle costs ~13x a tail cycle — and ~42% of that is the **evaluation & analysis** stage, where the AI scores every vendor's technical and commercial submission individually. That's the deepest, most defensible AI work in the platform, and it lands exactly where the procurement *value* is.

5 Adani at Scale — ~300,000 Cycles a Year

A group running ~300,000 procurement cycles annually has a heavily tail-weighted mix: a vast operational long tail, a small number of high-value strategic deals. Our working distribution:

TIER	SHARE OF VOLUME	CYCLES / YEAR	COST / CYCLE	ANNUAL AI COST
TAIL	50%	150,000	\$0.94	\$141,000
STANDARD	30%	90,000	\$3.04	\$273,600
TACTICAL	15%	45,000	\$4.56	\$205,200
COMPLEX	5%	15,000	\$12.26	\$183,900
Total	100%	300,000	\$2.68	\$803,700

6 The Headline Numbers

The estimated AI operating cost to run the full autonomous spine for a ~300,000-cycle-a-year group — the internal fixed cost that underpins the platform subscription.

Estimated annual AI cost

~\$804K

all agents - intake + pay - 300K cycles

Per month

~\$67K

≈ \$66,975 / month

Blended per cycle

\$2.68

across the full tier mix

Cheapest → costliest cycle

\$0.94 - \$12.26

tail → complex

✓ **What this buys**

~\$804K of AI a year runs **300,000 complete sourcing-to-pay cycles** end to end — every intake captured, every RFx drafted, every offer extracted, every bid scored, every award recommended, every invoice matched, with an always-on assistant on every screen. That is roughly **\$2.68 of AI per cycle** — a small fraction of the buyer-hours each cycle replaces.

7 What Actually Drives the Cost

• **Complexity, not volume**

The 20% of cycles that are tactical or complex drive ~48% of the bill. The 50% that are tail drive ~18%. Cost concentrates where the value is.

• **Evaluation is the single biggest line**

Per-vendor technical + commercial scoring on complex deals is the largest single cost — and the hardest work to do well.

• **The mix is the master dial**

Shift complex from 5% to 7% and the annual cost moves ~\$36K. The single best way to tighten this estimate is to measure the real tier distribution.

• **Costs trend down, levers exist**

Batching the heavy evaluation work and routing simpler tasks to the fast tier keeps the cost efficient — before any provider price drops.

8 Assumptions & How This Changes Over Time

Pricing basis	Current AI pricing. Heavy reasoning on the Deep tier; everyday drafting & evaluation on Balanced; high-volume extraction & ranking on Fast. Subject to change — historically downward.
Scope	AI processing only. Excludes platform subscription, hosting, database, search, and storage.
Volume	300,000 procurement cycles / year, group-wide.
Tier mix	50% tail - 30% standard - 15% tactical - 5% complex (group-scale, tail-weighted). The biggest single sensitivity.
Assistant usage	~2 help + ~1.5 analyst questions per cycle, amortised. Scales with user adoption.
Invoice base	~12 invoices per cycle (partial deliveries / multi-invoiced invoices).
Negotiation autonomy	Autonomous on tail/standard; buyer-led (AI-assisted) on tactical/complex.
Commercial treatment	Carried as Aerchain's internal fixed cost within the 3-year term subscription — not billed to Adani per token or per cycle. The customer platform price is fixed for the 3-year term regardless of movement in this internal cost (which historically trends down).
Refinement	We recommend instrumenting one live cycle per tier to replace estimates with measured numbers as the deployment beds in.
Status	Directional planning model — not a contractual quote.

The single message

Running Aerchain's full autonomous, intake-to-pay agent stack across **~300,000 cycles a year costs roughly \$804K in AI** at today's pricing — about **\$2.68 per cycle**. The number is dominated by the small share of complex, high-value deals, it falls as AI pricing falls, and we will re-baseline it against live usage. Adani pays a fixed platform subscription; this AI cost is carried internally by Aerchain.